

Data Mining for Business Analytics

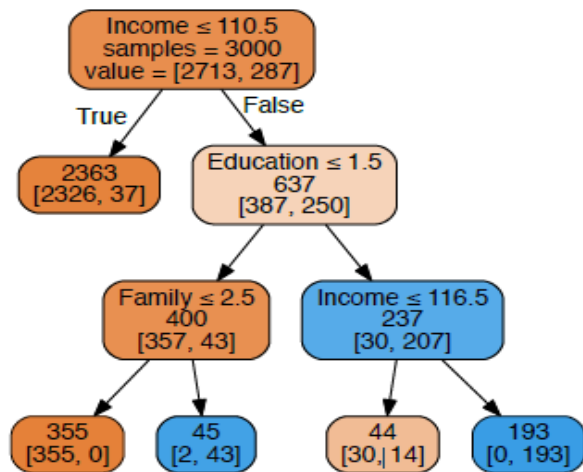
MSBA 511

Classification
Decision Trees

What is CART?

CART stands for **C**lassification **and** **R**egression **T**rees

- Classify or predict an outcome based on a set of predictor variables producing a tree structure



Output is a set of Rules

- Income > 110.5 **AND**
Education ≤ 1.5 **AND**
Family > 2.5

Classification Trees

Used when the target is **categorical**

Splits maximize homogeneity of classes in each branch

Regression Trees

Used when the target is **continuous**

Splits minimize variance within each branch

Key Features of Decision Trees

Binary Splits: Each split produces two child nodes

Pruning: Prevents a tree from being fully grown to minimize complexity and reduce overfitting

Non-Parametric: No assumptions about the underlying distribution of each variable

Interpretability: Trees are easy to understand and visualize

Many Real-World Applications

- Bank uses decision trees to classify transactions as risky for further review of fraud.
- e-Commerce company uses decision trees to classify customers into different segments (e.g., high-value, occasional shopper, one-time buyer).
- Healthcare company uses decision trees to predict if a patient has a specific medical diagnosis.
- Telecom company uses decision trees to classify customers as likely to churn for retention programs.
- Email provider uses decision trees to classify new emails as spam.

How is the Tree Produced

Recursive partitioning: Repeatedly split the records into two parts to achieve maximum homogeneity of outcome within each new part.

1. Pick one of the predictor variables, x_i
2. Pick a value of x_i , say s_i , that divides the data into two (not necessarily equal) portions
3. Measure how “pure” or homogeneous (containing records of mostly one class) each partition is
4. Algorithm tries different values of x_i , and s_i , to maximize purity in initial split
5. After you get a “maximum purity” split, repeat the process for a second split (on any variable)

Riding Mowers Example

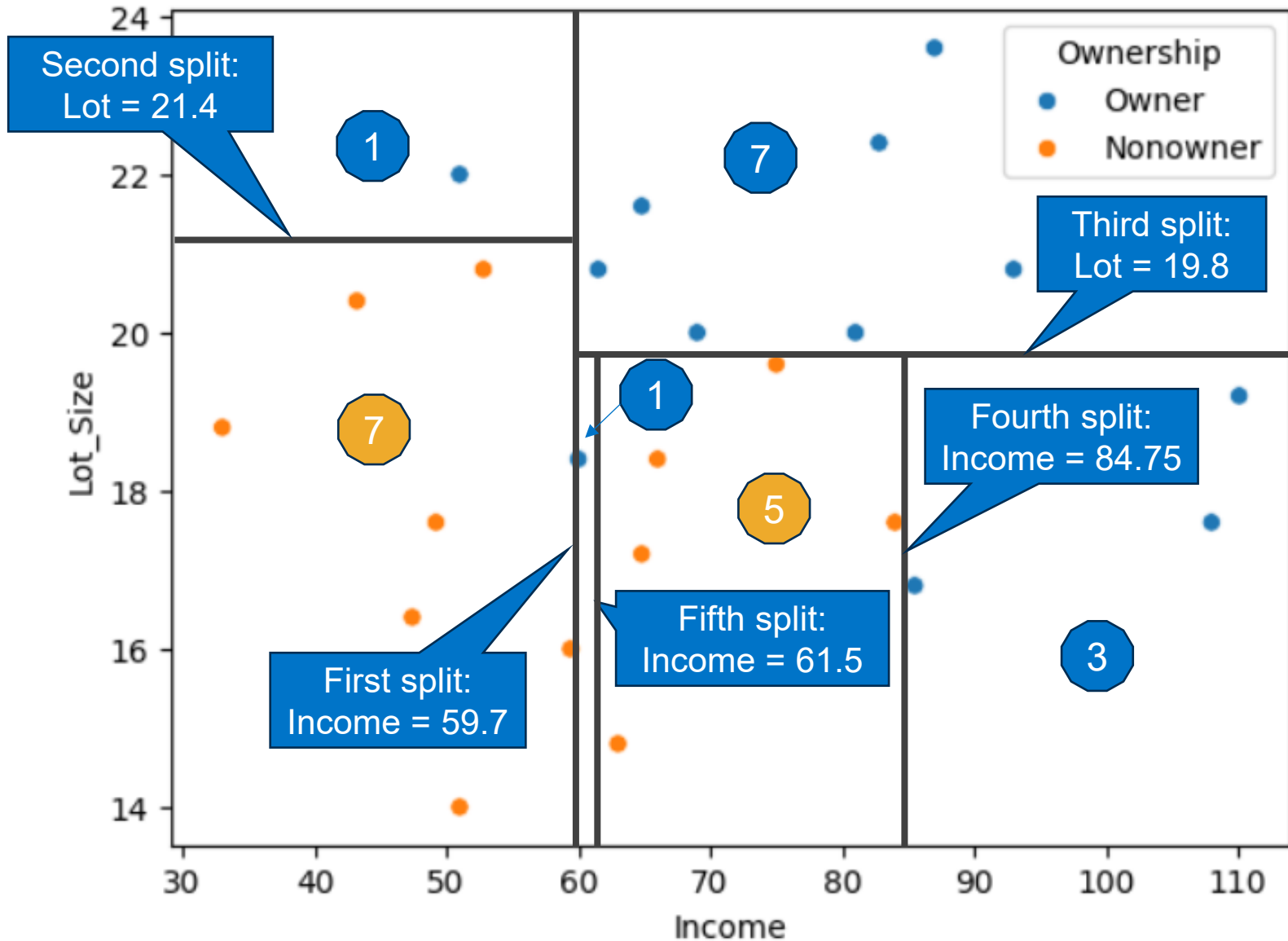
Income	Lot Size	Ownership
60.0	18.4	owner
85.5	16.8	owner
64.8	21.6	owner
61.5	20.8	owner
87.0	23.6	owner
110.1	19.2	owner
108.0	17.6	owner
82.8	22.4	owner
69.0	20.0	owner
93.0	20.8	owner
51.0	22.0	owner
81.0	20.0	owner
75.0	19.6	non-owner
52.8	20.8	non-owner
64.8	17.2	non-owner
43.2	20.4	non-owner
84.0	17.6	non-owner
49.2	17.6	non-owner
59.4	16.0	non-owner
66.0	18.4	non-owner
47.4	16.4	non-owner
33.0	18.8	non-owner
51.0	14.0	non-owner
63.0	14.8	non-owner

A company sells riding lawn mowers and is interested in **classifying** families in a city that are likely to purchase a riding mower. We have data for 24 households where **Ownership** of a mower is labeled. We have **predictor** variables **Income** and **Lot Size** to predict ownership.

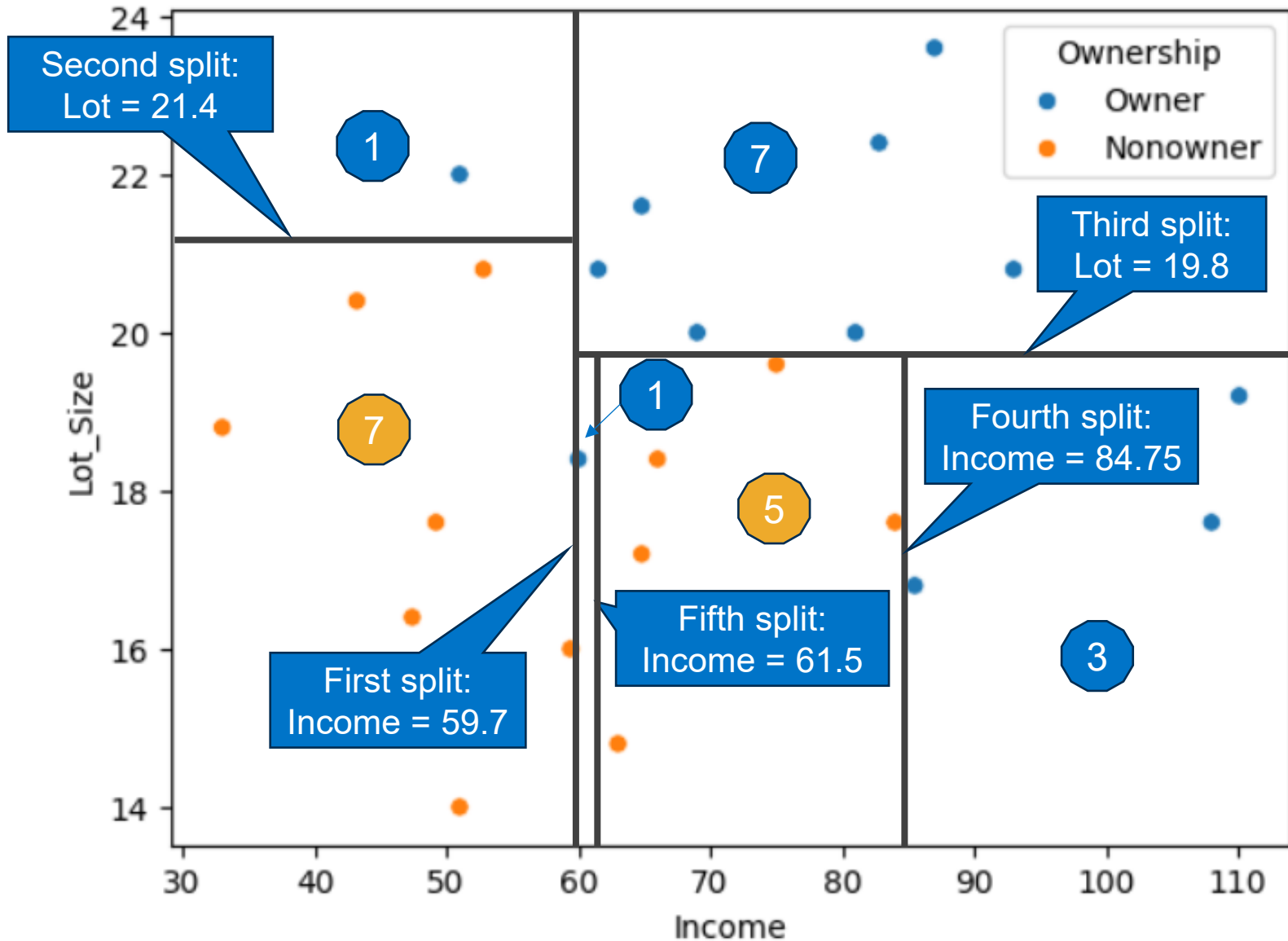
How to Split

- Order records according to one variable such as **Income**
- Take a predictor value, such as 59.7 and divide records into those with income > 59.7 and ≤ 59.7
- Measure resulting purity (homogeneity) of class in each node and then calculate the overall impurity (weighted average of each individual rectangle)
- Try all other split values
- Repeat for other variable(s)
- Select the one variable and split that yields the most purity

Visualize Splits



Any Concerns?



How to Measure Impurity

Gini Index

The Gini impurity index for rectangle A is defined by:

$$I(A) = 1 - \sum_{k=1}^m p_k^2$$

p = proportion of cases in rectangle A that belong to class k (out of m classes)

- **I(A)** = 0 when all cases belong to same class
- Max value = 0.50 when all classes are equally represented (binary case)

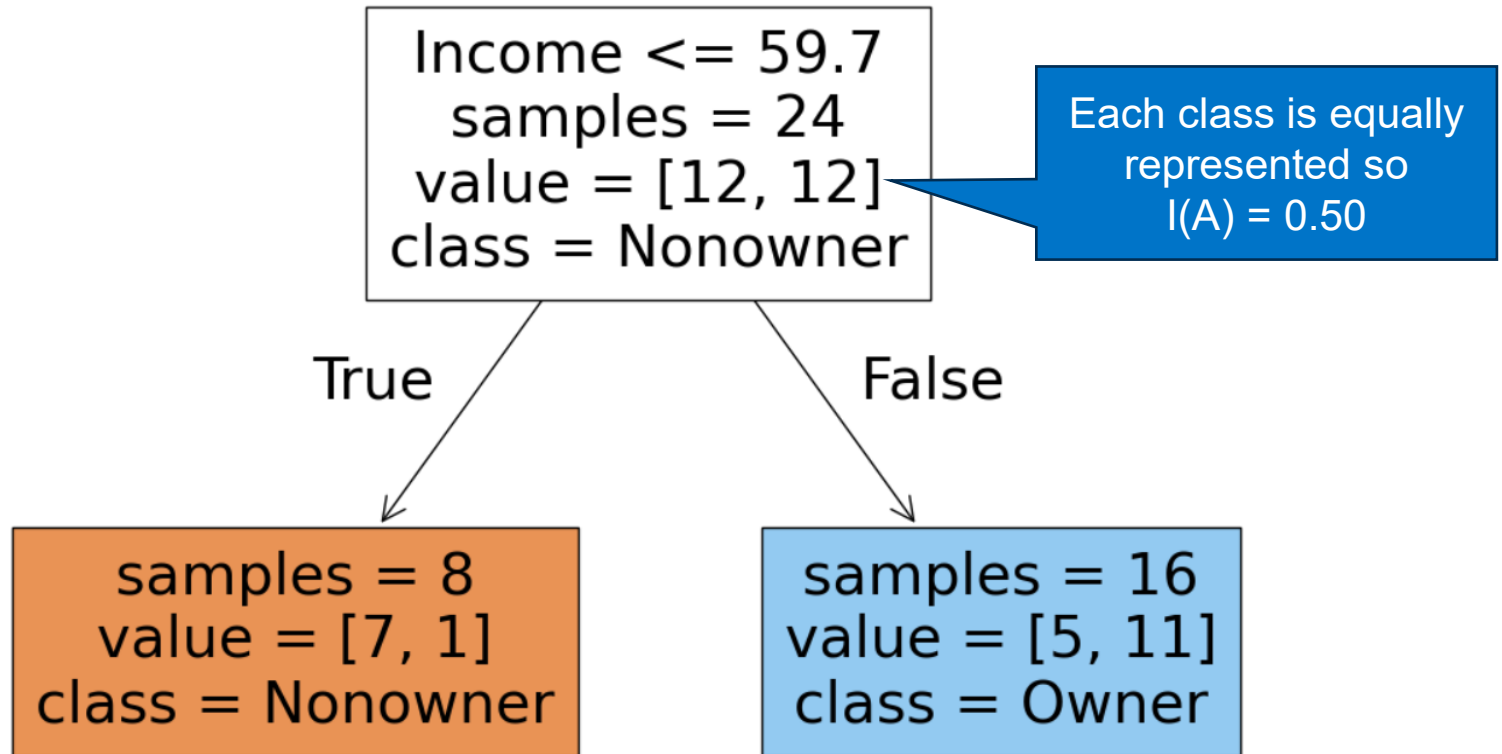
Entropy

The entropy for rectangle A is defined by:

$$\text{entropy}(A) = - \sum_{k=1}^m p_k \log_2(p_k)$$

- **Entropy** ranges between 0 (most pure) and $\log_2(m)$ (equal representation of classes)

First Split Gini



$$= 1 - \left(\frac{7}{8}\right)^2 - \left(\frac{1}{8}\right)^2$$

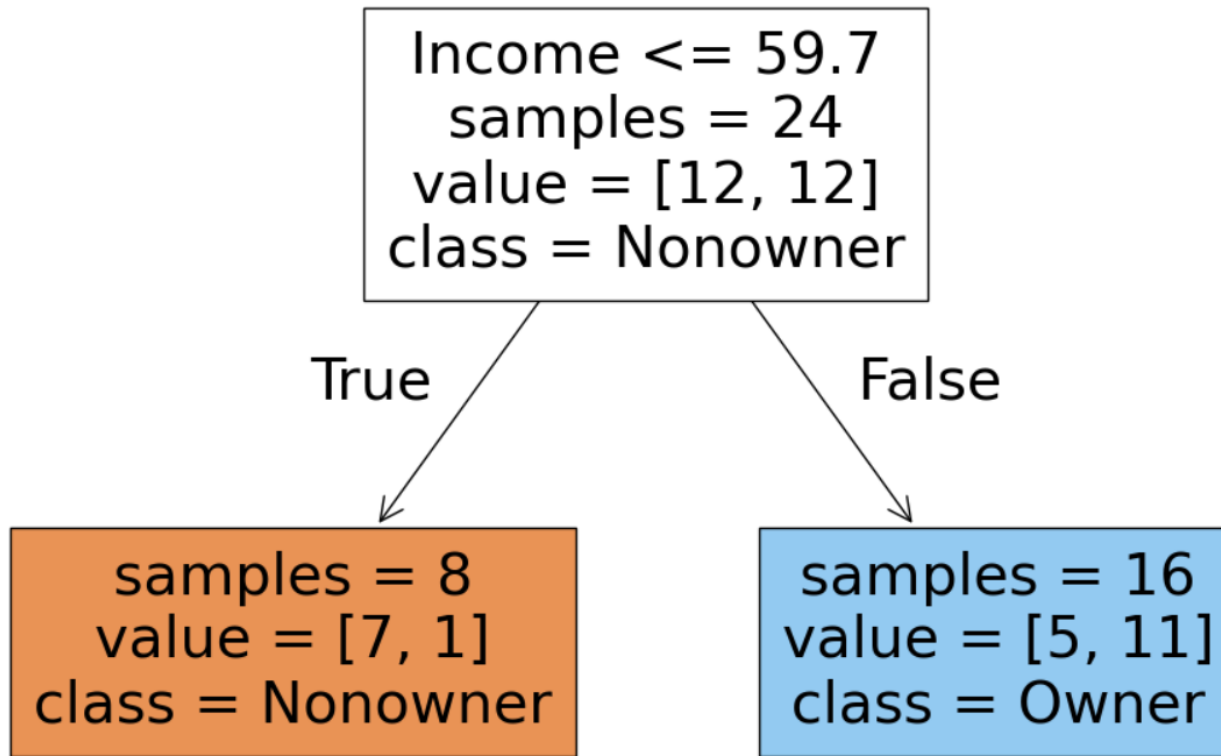
$$= \mathbf{0.219}$$

$$= 1 - \left(\frac{11}{16}\right)^2 - \left(\frac{5}{16}\right)^2$$

$$= \mathbf{0.430}$$

$$\text{Combined Impurity of the Two Nodes} = \left(\frac{8}{24}\right)(\mathbf{0.219}) + \left(\frac{16}{24}\right)(\mathbf{0.430}) = \mathbf{0.359}$$

First Split Entropy



$$= -\left(\frac{7}{8}\right) \log_2 \left(\frac{7}{8}\right) - \left(\frac{1}{8}\right) \log_2 \left(\frac{1}{8}\right)$$

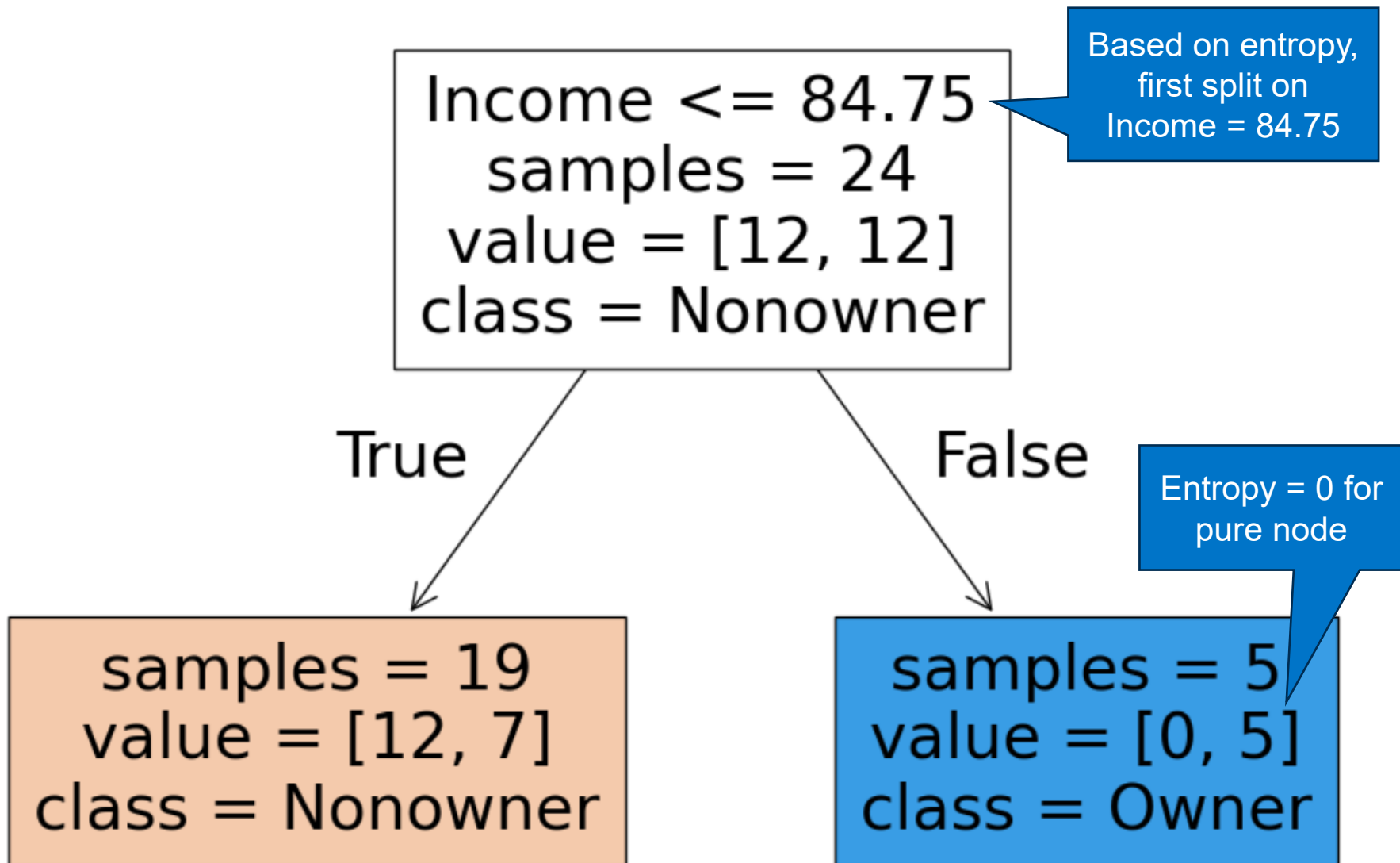
$$= \mathbf{0.544}$$

$$= -\left(\frac{11}{16}\right) \log_2 \left(\frac{11}{16}\right) - \left(\frac{5}{16}\right) \log_2 \left(\frac{5}{16}\right)$$

$$= \mathbf{0.896}$$

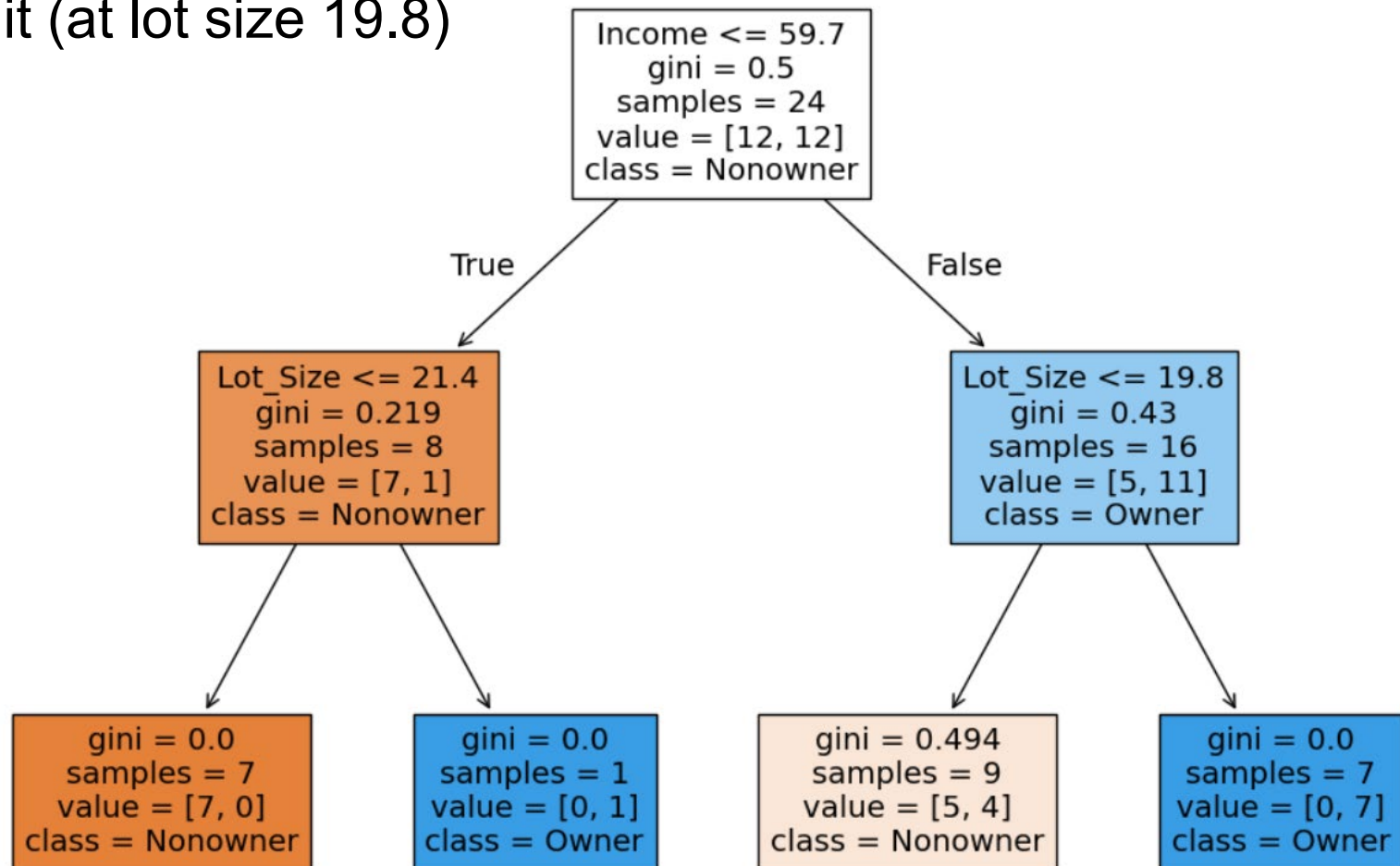
$$\text{Combined Impurity of the Two Nodes} = \left(\frac{8}{24}\right) (\mathbf{0.544}) + \left(\frac{16}{24}\right) (\mathbf{0.896}) = \mathbf{0.779}$$

Does the Choice of Impurity Measure Matter?

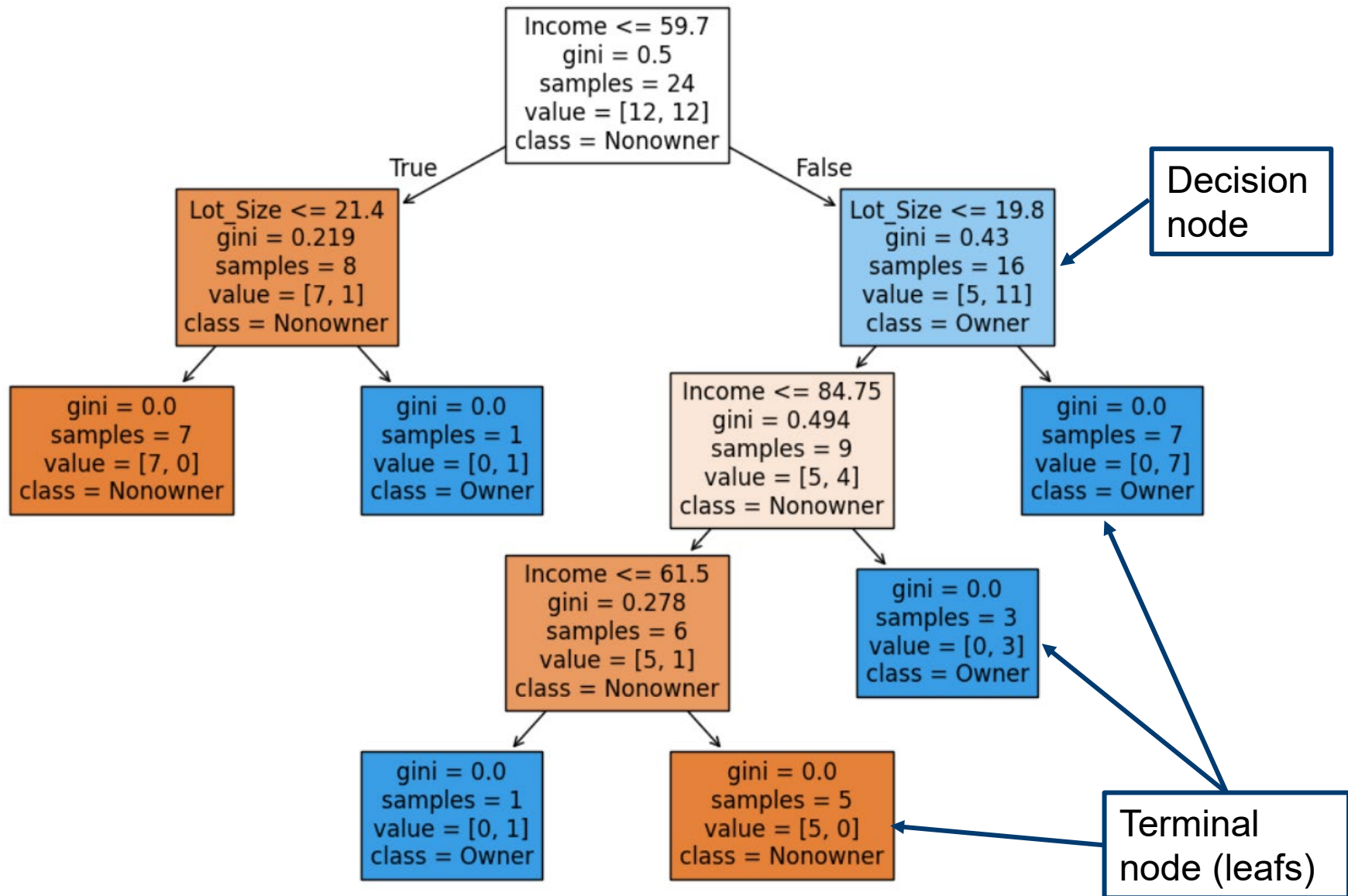


Tree After 3 Splits

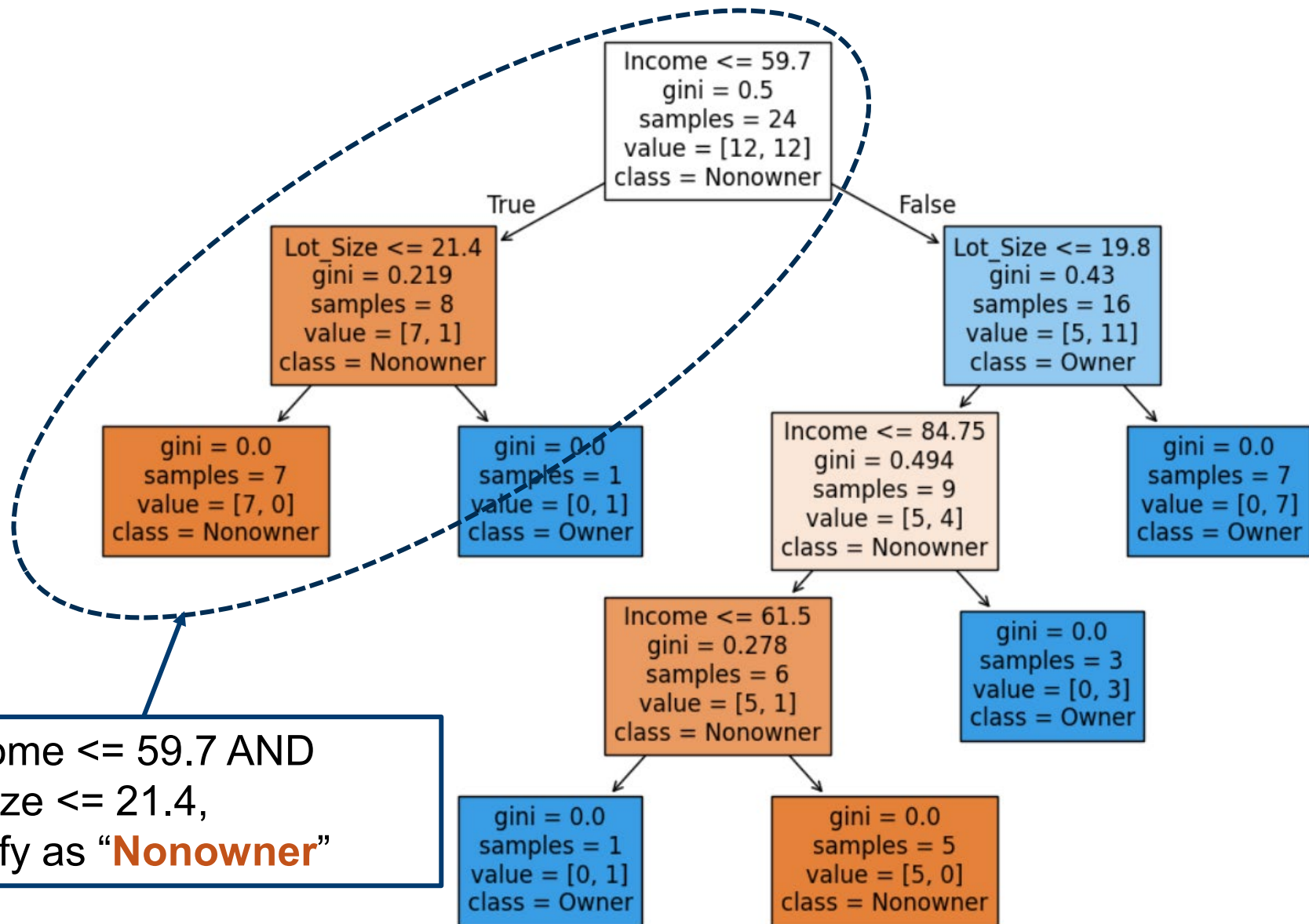
The first split is on Income, then the next split is on Lot Size for both the low-income group (at lot size 21.4) and the high income split (at lot size 19.8)



Tree After All Splits



Read Down to Derive Rules



Let's Practice: Personal Loan Example

In this example, we will work with only 3 binary variables to limit the number of options for the splits. The goal is to help a bank improve its conversion rate for personal loan acceptance in its next marketing campaign. We are working with a sample of 5000 customers and the predictor variables are whether the customer currently has a **Securities Account**, **CD Account** and **Credit Card** with the bank.

Here is summary of the data:

	Personal Loan	
Securities Account	No	Yes
No	4058	420
Yes	462	60

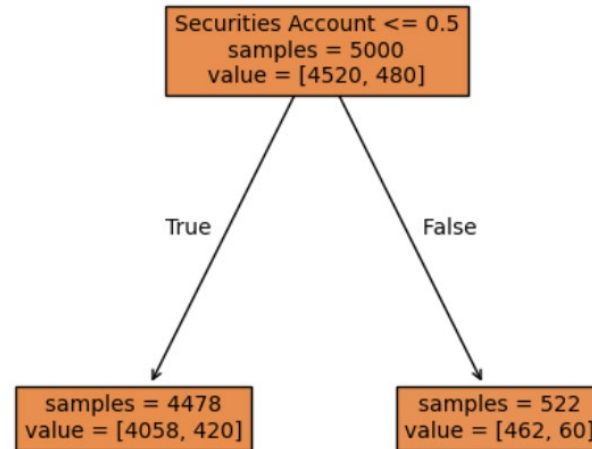
	Personal Loan	
CD Account	No	Yes
No	4358	340
Yes	162	140

	Personal Loan	
Credit Card	No	Yes
No	3193	337
Yes	1327	143

Let's Practice: Personal Loan Example

Calculate the combined impurity using the Gini index for **Securities Account**

	Personal Loan	
Securities Account	No	Yes
No	4058	420
Yes	462	60



$$\text{Securities Account} = \text{No} = 1 - \left(\frac{4058}{4478}\right)^2 - \left(\frac{420}{4478}\right)^2 = 0.170$$

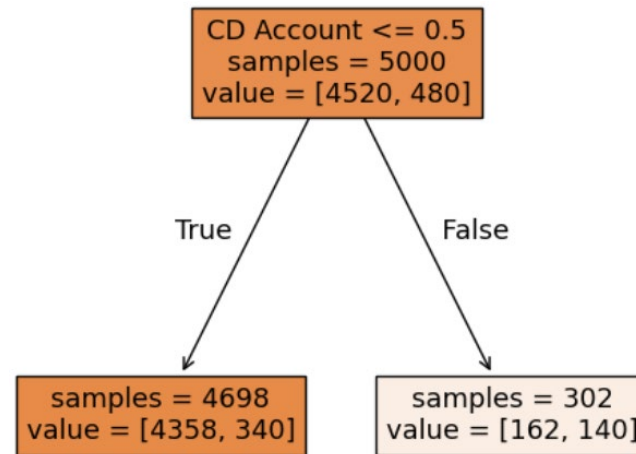
$$\text{Securities Account} = \text{Yes} = 1 - \left(\frac{462}{522}\right)^2 - \left(\frac{60}{522}\right)^2 = 0.203$$

$$\text{Combined Impurity of the Two Nodes} = \left(\frac{4478}{5000}\right)(0.170) + \left(\frac{522}{5000}\right)(0.203) = 0.173$$

Let's Practice: Personal Loan Example

Calculate the combined impurity using the Gini index for **CD Account**

	Personal Loan	
CD Account	No	Yes
No	4358	340
Yes	162	140



$$\text{CD Account} = \text{No} = 1 - \left(\frac{4358}{4698}\right)^2 - \left(\frac{340}{4698}\right)^2 = 0.134$$

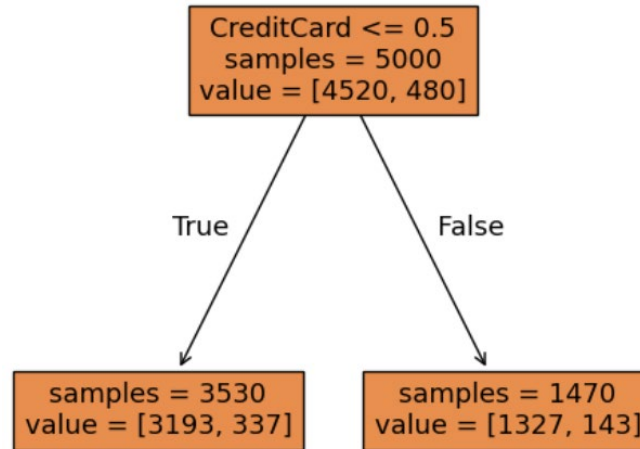
$$\text{CD Account} = \text{Yes} = 1 - \left(\frac{162}{302}\right)^2 - \left(\frac{140}{302}\right)^2 = 0.497$$

$$\text{Combined Impurity of the Two Nodes} = \left(\frac{4698}{5000}\right)(0.134) + \left(\frac{302}{5000}\right)(0.497) = 0.156$$

Let's Practice: Personal Loan Example

Calculate the combined impurity using the Gini index for **Credit Card**

	Personal Loan	
Credit Card	No	Yes
No	3193	337
Yes	1327	143



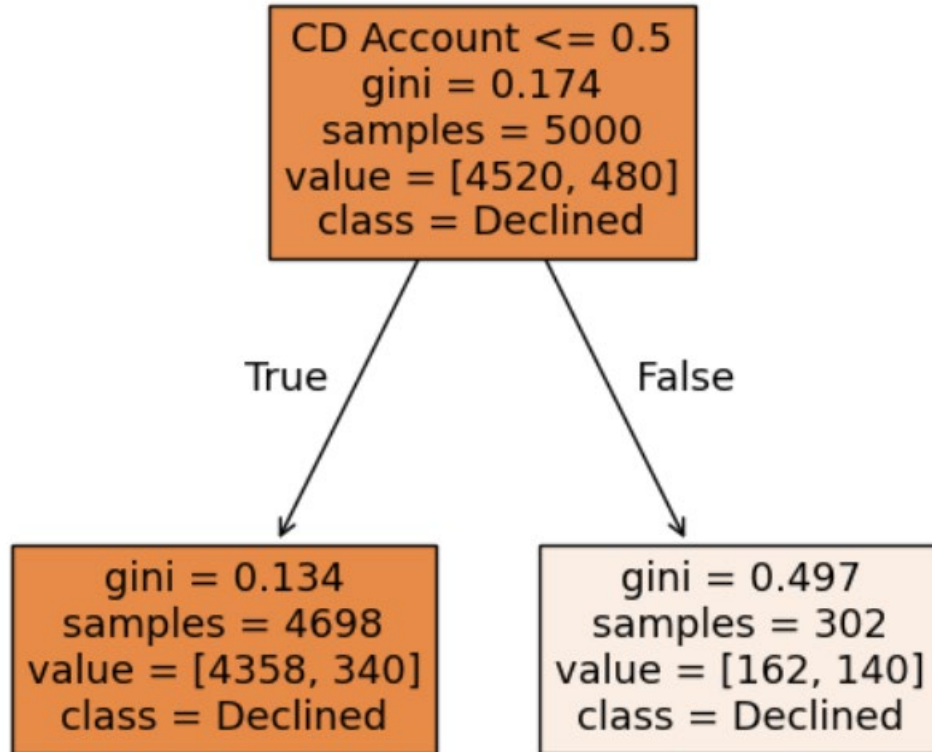
$$\text{Credit Card} = \text{No} = 1 - \left(\frac{3193}{3530}\right)^2 - \left(\frac{337}{3530}\right)^2 = 0.173$$

$$\text{Credit Card} = \text{Yes} = 1 - \left(\frac{1327}{1470}\right)^2 - \left(\frac{143}{1470}\right)^2 = 0.176$$

$$\text{Combined Impurity of the Two Nodes} = \left(\frac{3530}{5000}\right)(0.173) + \left(\frac{1470}{5000}\right)(0.176) = 0.174$$

Let's Practice: Personal Loan Example

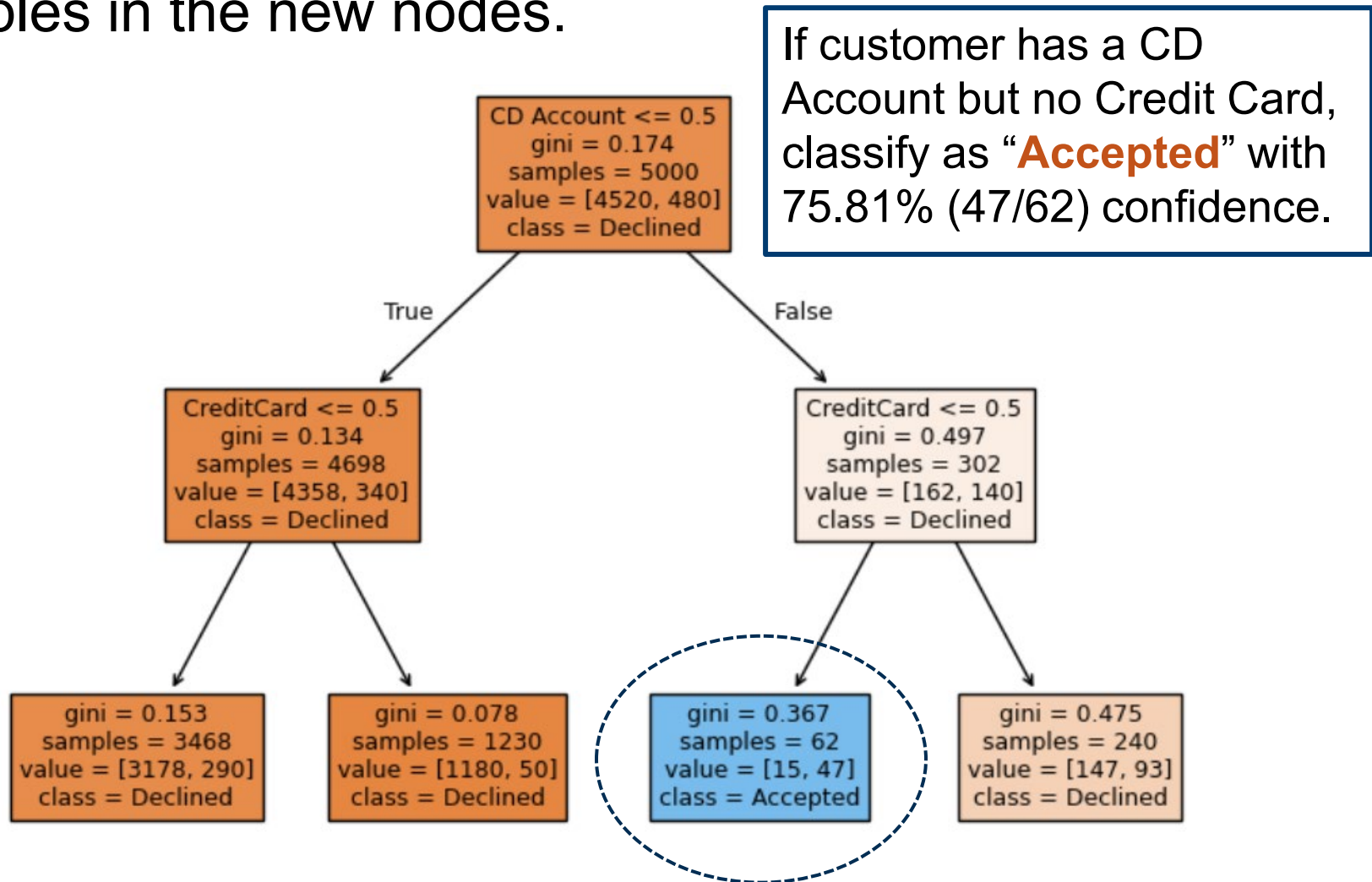
CD Account had the lowest impurity of **0.156** and is the first split!



Securities Account had the next lowest impurity of **0.173**. Should that be the next split?

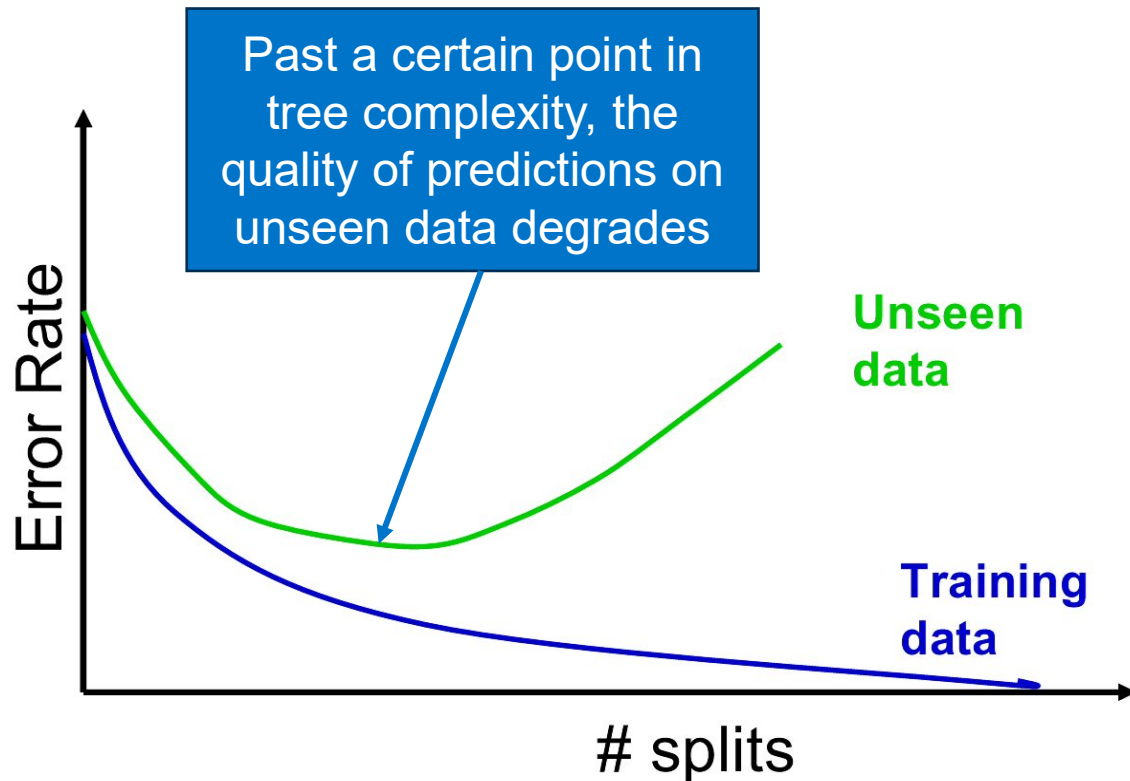
Let's Practice: Personal Loan Example

The next splits need to be evaluated based on the samples in the new nodes.



Overfitting Problem

- Natural end of process is 100% purity in each leaf
- This **overfits** the data, which end up fitting noise in the data



How to Avoid Overfitting

Pre-Pruning (Early Stopping)

- **Limit Maximum Depth:** Set a cap on the depth of the tree to prevent it from modeling noise.
- **Minimum Samples for Split:** Require a minimum number of samples to consider a split, ensuring that splits occur only when there's enough data.
- **Minimum Samples per Leaf:** Enforce a minimum number of samples at the leaf nodes to avoid creating overly specific branches.
- **Threshold for Splitting:** Establish a minimum improvement (e.g., minimum impurity decrease) before allowing a split.
- **Restrict Feature Usage:** Limit the number of features considered at each split to reduce the risk of capturing noise.

Post-Pruning

- **Cost-Complexity Pruning:** Grow the full tree first, then remove branches that provide little benefit using a complexity parameter.
- **Cross Validation:**
 - Evaluate the impact of removing branches and prune branches that do not improve performance.

Advantages and Disadvantages of Trees

Advantages:

- Easy to understand and to interpret.
 - Trees can be visualized to show the entire decision paths.
- Can handle both numerical and categorical data.
 - Please note the scikit-learn implementation does not support categorical variables.
- Does not require data normalization techniques.
- Able to handle categorical (binary and multi-classes) and numerical output.
- Not a black-box model so output is easily explainable.

Disadvantages:

- Can easily create over-complex trees that do not generalize well to unseen data.
- Can be unstable to small variations in the data that might result in a completely different tree structure.